

Overview

The `depthBarMermaid` helper was refactored to replace the old flowchart bar graph with an XY chart. The change trims the displayed depth slice from 14 to 12 entries and updates the empty-state message.

Key changes

- **Slice size** – `metrics.depthHistogram.slice(0, 12)` (line 189) replaces the previous `slice(0, 14)`.
- **Empty-state text** – `"No path depth data"` (line 191) replaces `"No data"`.
- **Graph type** – the function now returns an `xychart-beta` block (lines 197-203) instead of a `flowchart TB` with bar nodes (lines 193-197).
- **Label construction** – depth labels are generated via `labels = rows.map(d => \ d${d.depth})`` and `values = rows.map(d => d.count)`` (lines 193-196).
- **Y-axis scaling** – `yMax` is computed as ``Math.max(maxVal, Math.ceil(maxVal`
`1.15))`` (line 201) to give a 15% headroom.

Impact

- **Visual output:** Consumers of the Mermaid string will now see an XY chart instead of a bar graph, changing the appearance of depth histograms.
- **Performance:** Limiting to 12 rows reduces the amount of data rendered, slightly improving rendering speed.
- **Maintainability:** The new implementation is more declarative and removes manual node-generation logic, easing future tweaks.

Risks & follow-ups

- **Rendering failures:** Verify that the target environment's Mermaid version supports `xychart-beta`; older renderers may not support it, potentially breaking dashboards.
- **Test regressions:** Existing unit tests that assert the exact Mermaid string will fail; update expectations or add conditional checks.
- **Documentation:** UI or docs may need to explain the new chart type and its interpretation.
- **Edge cases:** Ensure that when `metrics.depthHistogram` has fewer than 12 entries, the chart still renders correctly and the Y-axis scaling behaves as intended.